WHITE PAPER · HWACHEON MEGA 95 × MILITARY & DEFENSE

Hwacheon Mega 95 for Military & Defense

Long-bed CNC lathe for extended shafts, rollers, rods, and structural round parts. This paper covers what the machine does, the military & defense parts we produce with it, the alloys we run, the tolerances and finishes military & defense buyers expect, and the documentation package.

16" OD × 120" L max turning ENVELOPE	**±0.0005" routine on critical features** TOLERANCE	**1994** SINCE
Same-week / rig-down direct RESPONSE		



Rattle & Hum · 16" OD × 120" L max turning · long-bed heavy-duty CNC lathe

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Executive summary

Machine: Hwacheon Mega 95 (shop nickname: *Rattle & Hum*). long-bed heavy-duty CNC lathe at B&R Productions, running from our New Waverly, Texas shop.

Application: Long-bed CNC lathe for extended shafts, rollers, rods, and structural round parts for Military & Defense customers — Defense primes, subassembly manufacturers, tactical vehicle component OEMs, munitions component suppliers, weapon-system integrators, sustainment (out-of-production replacement) contractors.

Envelope: 16" OD × 120" L max turning. **Tolerance:** ±0.0005" routine on critical features. **Traceability:** Material by heat and lot on every job.

The machine, in context

Ten feet of turning length is a specialty capability. Long drive shafts, extended mandrels, rig-floor rollers, oil-and-gas structural rods — the parts that don't fit on a 4- or 6-foot lathe. Rattle & Hum lets us keep this class of work in-house instead of outsourcing the finish turning after roughing on a larger machine somewhere else.

Why Rattle & Hum for Defense work

Long defense round parts — shafting, rods, long structural rounds up to 120 inches, single piece. On ground-vehicle and equipment hardware, eliminating a coupling eliminates a wear point and an alignment adjustment that somebody has to make in the field.

What this looks like on a real part

A long drive or torsion shaft is a straightness and concentricity part along its whole length, not just at the journals. At ten feet the stock sags under its own weight before the tool ever touches it, so support staging is decided before the first cut. Roughing heat then grows the part measurably; finishing a hot long shaft gives you something that gauges correctly at the machine and is bowed by the next morning.

> **When this is the wrong machine**
>
> Non-controlled work only. 16" OD ceiling; larger diameters go to Big Thicket Express. No through-bore — long tubular work goes to Texaco Thunder.

Full specifications

Model	Hwacheon Mega 95 ("Rattle & Hum")

Type	Heavy-duty gap-bed CNC lathe — Mega-95 platform
Max turning capacity	16" OD × 120" L (B&R working spec)
Platform max swing over bed	38.5"
Swing through gap	48.4"
Swing over cross slide	26.2"
Distance between centers	10 ft (3,000 mm) B&R config; platform available to 205"
X-axis travel	19.09"
Spindle bore	6"
Spindle speed	40 – 1,000 RPM (heavy-duty range)
Spindle motor	30 HP
Chuck	24" 3-jaw hydraulic
Turret	12-position
Control	Fanuc 21-T (typical)
Machine weight	~22,000 lb
Tolerance capability	±0.0005" routine on critical features; straightness held over full length
Materials	Full oilfield alloy range — Inconel, Duplex, 17-4 PH, 4140/4340, Monel

Values marked "platform" reflect the machine manufacturer's published spec range for this platform family; B&R's working spec is what we quote on this specific unit.

Who this serves in Military & Defense

Typical buyer: Defense primes, subassembly manufacturers, tactical vehicle component OEMs, munitions component suppliers, weapon-system integrators, sustainment (out-of-production replacement) contractors.

What's hard about military & defense work: Sensitive prints, program-specific quality plans, tight-tolerance structural components, exotic-alloy selection driven by service environment, and legacy-part sustainment where the OEM no longer supports the design and the prime's supply chain has moved on.

Typical Military & Defense parts we produce on Rattle & Hum

- Extended structural rods and shafts
- Long weapon-system components
- Extended defense-platform structural parts
- Long-form actuator components
- Sustainment parts requiring 8+ foot turning length

Above list is representative; if your part isn't shown, call and ask — most oilfield / aerospace / defense / industrial turned or milled work fits somewhere in the shop.

Alloys we run for Military & Defense

Standard range: 17-4 PH · 15-5 PH · 13-8 Mo · 4140 pre-hard · 4340 · Titanium Grade 5 · Inconel 718 · Aluminum 7075 · High-strength steels · Customer-specific alloys on request.

Defense parts routinely spec 4140 pre-hard, 4340, and precipitation-hardening stainless — condition-specific machining knowledge matters, same as aerospace. Titanium is common on weight-critical structural components. Legacy sustainment often specifies materials nobody sees anywhere else; we work with the spec sheet you provide and validate machining strategy before we cut.

> **What "we run these weekly" means:** we stock or reliably source the common alloys in the industry's typical spec range. Sharp tooling, proven feeds and speeds, and process control specific to each alloy family. Your part is not the one we're experimenting on.

How we machine it — our 5-step process

This is the process B&R runs on every job, military & defense or otherwise. The specifics of feeds, speeds, and tooling shift by alloy and machine; the discipline is universal.

1. **Material verification.** Every job starts with material traceability. Heat and lot numbers documented against the mill test report. For customer-supplied material, we verify against the CofC before touching the stock.

2. **First-article layout.** Stock is measured and laid out before any cutting starts. Concentricity, roundness, and squareness of the starting material established; if the stock is out of tolerance we call the customer before wasting time on a bad blank.

3. **Roughing with process control for the alloy.** Feeds, speeds, and coolant strategy set for the specific alloy family — sharp coated carbide (or PCD for high-volume Ti Grade 5), positive rake, high-pressure through-tool coolant on 2507 and Inconel to control cutting temperature. Interrupted cuts get extra attention because they cycle heat into the workpiece.

4. **Finish pass to spec.** Finish pass with a fresh insert, controlled DOC, and measured surface-finish targets. Critical features (bore concentricity, seat grooves, sealing surfaces) are held tighter than most job-shop lathes can offer.

5. **CMM verification + documentation.** Every critical dimension CMM-verified. First-article report generated. Certificate of conformance issued with delivery. If your quality group needs a documentation package beyond that, tell us with the RFQ and we'll be straight about what we can and can't produce in-house.

Tolerance and finish expectations in Military & Defense

±0.0005" routine on critical features. Tighter with proper setup and fixture. GD&T-driven tolerancing supported. CMM verification on first articles; customer-plan-driven measurement on production runs. Documentation the level a program office expects.

Documentation and quality workflow

Sensitive prints handled under NDA with document control — prints stay in-shop and aren't sent to outside vendors. First-article documentation standard. Material traceability by heat and lot on every job. Certificates of conformance with delivery. One thing we state plainly up front: B&R is not ITAR-registered with the U.S. State Department. If your print is ITAR-controlled, we are not the right shop and we'll tell you that on the first call. Non-controlled defense-adjacent work — commercial-spec components, sustainment parts, tooling, and fixtures — we quote all day.

Response time and emergency work

Program-driven schedules honored. Emergency component replacement supported for sustainment programs with established relationships. Legacy-part reverse-engineering supported for out-of-production sustainment.

How we compare

Compared to a large defense-prime captive shop: we're faster for one-offs, small runs, and legacy sustainment where the prime's supply chain has moved on. Compared to a lowest-bidder commodity shop: we understand the alloys, we document properly, and we don't guess at a condition-specific machining plan. Compared to an ITAR-registered shop: for controlled prints they win, because that's a State Department registration and we don't hold it. Know which bucket your part is in before you shop it — it saves everyone a wasted quote cycle.

Working with B&R Productions

New customer or existing, the process starts the same way: call (936) 291-7827 or send a print through the quote form. Prints are accepted as PDF, STEP, IGES, DWG, or DXF; if you only have a sample part, we reverse-engineer routinely. NDAs on request, and prints stay in-shop rather than going out to third-party vendors. Note for defense and aerospace buyers: B&R is not ITAR- or AS9100-registered, so export-controlled prints and AS9100 flow-down programs need a registered shop — we'll tell you that on the first call, not after the PO.

Quotes typically come back within 24 hours for standard work, same-day for repeat customers on stocked materials, and near-immediately by phone for emergency and rig-down situations. Lead times are quoted honestly — we don't promise Wednesday and deliver next month.

Frequently asked questions

8 questions covering the Hwacheon Mega 95 platform and Military & Defense work. Also indexed as FAQ schema for AI answer engines.

What's the Mega 95's working envelope?

16" OD × 120" L (10 ft) is B&R's working spec. Platform max swing is 38.5" over bed with a 48.4" swing through the gap and 26.2" swing over cross slide.

What kind of parts does the Mega 95 run?

Long shafts, extended mandrels, rig-floor rollers, drive rods, and oil-and-gas structural round parts — anything that doesn't fit on a 4- or 6-foot lathe. The 24" 3-jaw hydraulic chuck accepts big roughing blanks.

What's the spindle bore and speed range?

6" spindle bore, 40–1,000 RPM. 30 HP motor. Control is Fanuc 21-T on our unit; machine weight ~22,000 lb.

Do you handle ITAR-controlled defense work?

No — and we'd rather be the shop that tells you that up front. ITAR compliance is a registration with the U.S. State Department's DDTC, not a set of practices a shop can simply adopt, and B&R does not hold it. If your print is ITAR-controlled, you need a registered facility. What we do handle: non-controlled defense-adjacent work — commercial-spec components, legacy sustainment parts, tooling, fixtures, and reverse-engineered replacements for programs the OEM walked away from. Sensitive prints stay in-shop under NDA with document control, and first-article documentation and material traceability come standard.

What kind of defense-adjacent parts is this shop suited to?

Structural and mounting hardware, housings and enclosures, hydraulic components, ground-vehicle and equipment parts, tooling and fixtures, and reverse-engineered legacy parts for platforms the OEM no longer supports — all on non-export-controlled work. That's a description of what the equipment and the alloy experience suit, not a claim about specific programs. If you want to know whether we've run something like your part, ask and we'll tell you straight.

What alloys are typical for defense work at B&R?

17-4 PH, 15-5 PH, 4140 pre-hard, 4340, Titanium Grade 5, Inconel 718, aluminum 7075, high-strength steels. Customer-specific alloys on request.

Can you reverse-engineer legacy defense parts?

Yes — reverse-engineering from a worn sample is core job-shop work and it's what a shop like this is for. Send the sample and any documentation that survives; we'll measure it, verify the alloy, and produce the replacement with full material traceability.

What tolerances do you hold on defense components?

±0.0005" routine on critical features. Tighter with the right fixture. Every critical dimension CMM-verified. Documentation to customer spec.

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Ready to talk about your Military & Defense work?

Send us a print or a sample photo and specs. Emergencies: call direct and say "rig-down."

Published by B&R Productions — a precision CNC machining shop in New Waverly, Texas, in business since 1994. Serving oil & gas, aerospace, defense, and industrial customers across Texas and the Gulf Coast.

Written by the B&R Productions team. Machine specs verified against manufacturer data. Alloy and process notes reflect shop practice as of 2026-01-15.